Hackachusetts 2026

Project research: what we found and what we recommend

Saturday 24 October 2026 · Build day: 09:00–20:00 (about 11 hours)

What this document is. One place to review all the research: what the event actually asks for, the datasets it gives us, how ideas were picked and dropped, the top five ideas with build plans, and the decisions the team has to make tonight.
How much to trust it. Every dataset marked **LIVE** was fetched by `curl` on 24 Oct 2026 and returned real data. Items marked **listed only** appear in the official portal but were *not* fetched. URLs in the evidence tables from earlier rounds were spot-checked; quotes labelled *verified* were seen word for word. Anything else should be checked before you rely on it.
Source of truth. The workspace `~/research/` (`00_INDEX.md`, `40_shortlist.md`, `90_log.md`, `20_findings/`, `30_candidates/`). This PDF is a readable summary of those files.

Prepared for a 6-person team: 4 CS undergrads, 1 math grad student (spec writer), 1 chemistry major. Laptops only, free public data, one LLM coding agent.

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1 Executive summary

Bottom line. Earlier research used the wrong track list and never used the official data portal. Under the real tracks, the earlier top three (Mass Save rebates, FRC mentor logging, FRC inspection) fall out. The new top five are built on the portal's datasets, each one maps to a named challenge, and together they cover all four tracks.

#	Idea	Track	I	F	N	Why it ranks here
1	**311 triage**: repeat reports & missed deadlines in Boston 311, with an equity overlay	Civic	4	4	4	Uses the exact columns the challenge names; real city data; a resident can actually use the demo.
2	**Bridge triage**: maintenance priority list from MA bridge-inspection data	Robotics	3	4	4	Matches the challenge wording (the bottleneck is now interpretation); clean static file.
3	**Path-planner classifier**: which algorithm to use for a given map	Robotics	3	5	4	Safest build (pure Python, offline). Backup if a data-heavy idea stalls.
4	**Emergency outreach**: which neighbourhoods get missed in a flood	Civic	4	4	4	Strong equity angle on 3 live datasets; has a reliable fallback mode.
5	**Trial finder**: MA clinical trials in plain language, by distance	Health	3	4	3	Clean, safe build on a live federal feed; gives us a Healthcare entry.

I = impact, F = feasibility in 11 h, N = novelty / how under-served the problem is. All scores are 1–5.

Beyond the top five: Section 7 gives **8 more ideas** (an MBTA step-free route checker, a robot-data linter, swarm search-and-rescue, a MassHealth notice decoder, and others). They are less researched, but several have the same live data behind them.

Decisions needed tonight (details in Section 11):

1. Choose **one** idea, or two at most. With 6 people and 11 hours, two ideas is already a stretch.
2. If it is #1: download a Boston 311 data slice *tonight*. The city site blocks server-side downloads (Cloudflare).
3. Install-test the libraries on every laptop tonight (PythonRobotics, Leaflet, pandas).
4. Decide whether to focus on one track (Civic is the most open) or spread across tracks.

2 What the event actually asks for

2.1 Where this comes from

The site `https://hackachusetts2026.vercel.app/` renders in the browser, so normal page fetchers return a blank page. We downloaded its JavaScript bundle and pulled out the track, challenge, and dataset definitions directly. The event's own guidance says: *"These are not the final ideas on each track. They are just for guidance."* Two pages linked from the site (the coalition event page and a Notion rules page) also render in the browser and could **not** be read. **Read the judging criteria**

and team rules in a browser tonight.

2.2 The four real tracks

Track	Official one-liner	Correction to earlier research
Health — Healthcare & biotech	Turn continuous data, coverage, and research into better care.	Roughly matches the old "Healthtech/Biotech". Note that it puts weight on wearables, MassHealth coverage, and access, not only lab biotech.
Robotics — Robotics & hardware	Bring intelligence into the physical world.	The challenges are about physical AI: automation, simulation, data quality, inspection, planning. Youth-team logistics (the FRC ideas) do not fit this.
Climate — Energy & climate tech	Tackle how we power our world and protect our planet.	Was "Energy". Climate resilience, the ocean, and the coastal economy are also in scope.
Civic — Civic action	Strengthen the places, communities, and opportunities that shape daily life.	**New.** There is **no Education track**; education and college ideas belong here.

2.3 All 20 challenge statements, and which ideas address them

Columns: **1**=311 triage, **2**=bridge triage, **3**=path planner, **4**=emergency outreach, **5**=trial finder, **P**=a parked idea covers it. Empty rows are challenges *nobody on our list addresses yet.* Those are open opportunities, but they were not researched this session.

Track	Challenge (summary of the official text)	1	2	3	4	5	P
Civic	**Better Together**: everyday problems with housing, transport, public space, community, opportunity						
Civic	**Citizen-Friendly Digital Government**: complex websites, long PDFs, language barriers, confusing applications						●
Civic	**More Responsive Neighborhood Services**: 311 volume, repeat reports, prioritizing, reporting progress	●					
Civic	**Stronger Community Emergency Readiness**: fast-changing information; vulnerable people are hardest to reach				●		

Track	Challenge (summary of the official text)	1	2	3	4	5	P
Climate	**Resilient communities in a changing climate**: heat, flooding, storms; risk differs by neighbourhood				•		•
Climate	**Cleaner, more affordable buildings**: old buildings, retrofit cost, uneven program access						•
Climate	**A more reliable clean energy system**: peaks, severe weather, adding renewables, cost						•
Climate	**Understanding and protecting the ocean**: combining vessel, buoy, glider, and satellite data						
Climate	**Sustainable blue economy**: fishing, aquaculture, shipping, offshore wind sharing the same water						
Health	**Turning Wearable Data into Better Healthcare**: continuous monitoring into useful insight						
Health	**Keeping Residents Connected to Health Coverage**: MassHealth renewal, missed notices, upcoming rule changes						
Health	**Improving Healthcare Access and Coordination**: referrals and scheduling; rural travel distance						
Health	**Accelerating Biotechnology Innovation**: AI for research and experiments						•
Health	**Making Clinical Research More Accessible**: finding trials; technical eligibility text; language, distance					•	
Robotics	**Making Automation Accessible to Manufacturers**: small shops lack the staff and capital						
Robotics	**Robo-Gym Training Environments**: simulated real places for physical-AI training						
Robotics	**Real-Time Control for Data Quality**: catching bad robot-training data during collection						
Robotics	**Swarm Robotics for Search-and-Rescue**: coordination, mapping, task allocation						
Robotics	**Safer Inspection of Critical Infrastructure**: MassDOT already flies drones; interpreting what they capture is now the bottleneck		•				
Robotics	**Path Planning Classifier**: pick a planning algorithm for the environment and compute budget			•			

The portal also groups civic datasets under five extra headings: *Simplified public services, Responsive*

and resilient communities, Safer and connected places, Better college experiences, Pathways to education and opportunity. The last two are where former "education" ideas would go.

3 How ideas were selected

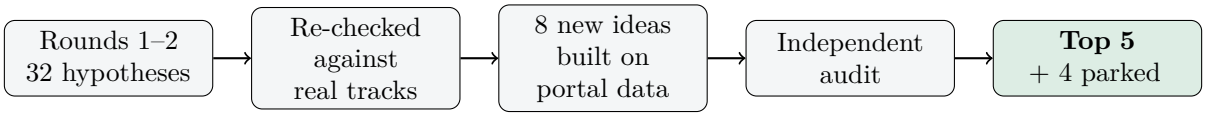
3.1 What changed from earlier rounds

1. **Wrong tracks.** Rounds 1–2 assumed Healthtech, Robotics, Energy, Education.
2. **Ignored the portal.** The event provides 166 datasets and resources, chosen for each challenge. None of the earlier top picks used any of them. In a hackathon built around a data portal, judges will look for honest use of that data.
3. **Too focused on single individuals.** Earlier gates required a dated forum post from one named person and dropped any idea with 3 or more competitors. Following your steer, the target is now *real problems that real people and organizations have, which existing tools solve poorly or not at all.* Evidence can be a government program, a documented structural gap, or the challenge text itself.

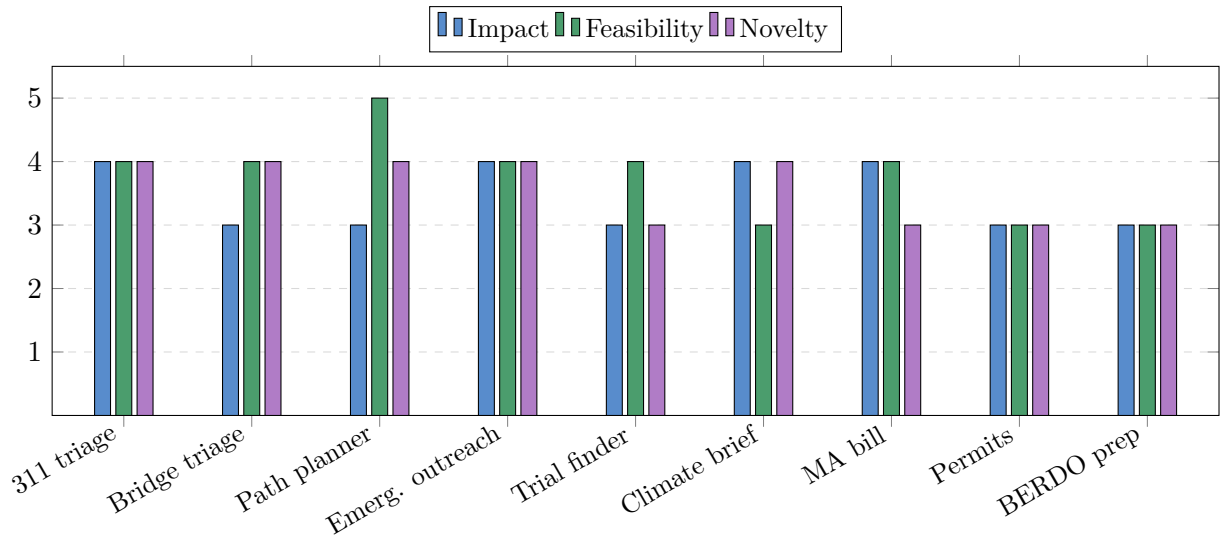
3.2 Gates: old vs. new

Gate	Rounds 1–2	Round 3
Real	Dated first-person complaint required	Documented real-world problem: challenge text, government program, data, or practitioner traces
Buildable	11 h, laptops, LLM agent	Same, plus the data must download from *our* laptops tonight
Under-served	3+ competitors means dropped	Existing tools serve it *poorly*; competition is a warning, not an automatic drop
Track fit	Old 4 tracks	Real 4 tracks **and** a named challenge
Uses portal	(not checked)	**New:** provided MA datasets do the real work in the build, not just decoration
Safe	No health PII, no advice, etc.	Same; also no auto-written official messages
Demo	A 60-second before/after	Same, plus a cached fallback so the demo can't hang

3.3 Pipeline at a glance

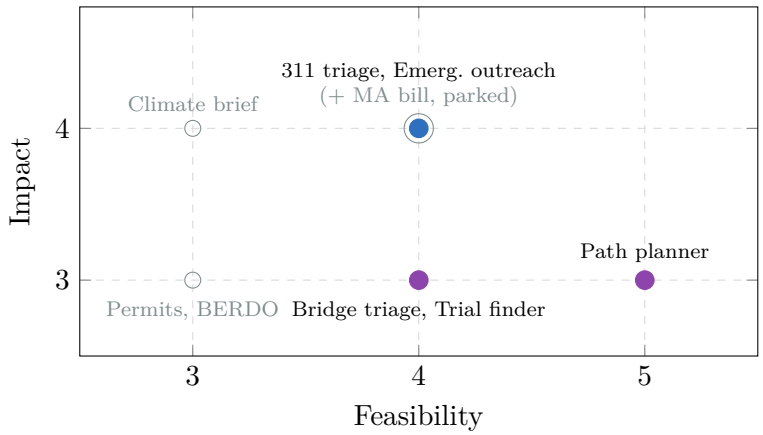


4 Score comparison



The first five are the shortlist; the last four are parked. Scores are judgment calls from the evidence, not measurements. Ties are broken by how much the idea relies on portal data and how reliable its demo is.

Impact vs. feasibility (shortlist in colour)



5 The shortlist: full idea cards

#1 311 triage — repeat reports and missed deadlines, with an equity overlay

Track / challenge Civic — More Responsive Neighborhood Services (311)
Scores I4 F4 N4
Problem The challenge itself says: *"high volumes and repeated reports can make it hard to recognize patterns, prioritize action, and communicate progress."* Duplicate reports of the same pothole inflate the counts and hide problems that never get fixed. Residents cannot tell whether their block is routinely ignored. Typical 311 dashboards show volume. They do not merge

duplicates or show where missed deadlines cluster.

Who it helps City operations staff (prioritizing work) and residents and councillors (seeing chronic neglect).

Data

- Boston 311 requests: `service_name`, `case_status`, `on_time`, `neighborhood`, `ward`, `open_date` — **BLOCKED** for server-side `curl` (Cloudflare). Should work from a browser. **Download a slice tonight.**
- MassGIS environmental-justice block groups (minority %, limited-English %, median income) — **LIVE**.

Demo (60 s) Pick a neighbourhood. A map shows open cases and cases that missed their deadline. Duplicate pothole reports collapse into one hot spot. The top three chronic categories are listed. An overlay shows that missed deadlines concentrate in a limited-English block group.

Hardest steps (1) Getting the data past Cloudflare. (2) A duplicate rule (same service, nearby, close in time) that doesn't merge unrelated reports. (3) An equity claim that holds up after normalizing for report volume.

Build plan

09:00	Math grad writes the spec: duplicate rule, "missed" means `on_time` is false, per-capita normalization. Data slice must already be cached.
10:00	CS1+CS2: load data and cluster duplicates. CS3: Leaflet map. CS4: join to EJ block groups.
13:00	Chem major (tester): check clusters and counts against the raw rows.
15:00	Chronic-category ranking and a one-line summary for residents.
16:30	Rehearse the demo story. **18:00 freeze** on cached data.

Risks / fallback If the live download fails, the cached slice is the demo (plan for this from the start). If the duplicate rule merges wrong reports, show the raw rows behind every cluster. If the equity pattern isn't there, report what the data actually shows; that is still a useful tool.

#2 Bridge triage — maintenance priorities from MA bridge-inspection data

Track / challenge Robotics — Safer Inspection of Critical Infrastructure

Scores I3 F4 N4

Problem The challenge says MassDOT Aeronautics *already flies drones* to inspect bridges, and *"the bottleneck has moved"* from capturing data to interpreting it. The National Bridge Inventory (NBI) is a wall of coded columns. A ranked, explained "which bridges need attention, and why" list is the interpretation step the challenge asks for.

Who it helps State and municipal bridge asset managers and inspection planners.

Data

- NBI Massachusetts file (`MA22.txt`) — **LIVE**, real condition columns.
- FHWA InfoBridge and the new SNBI schema docs — **listed only** (needed to decode the columns).
- dacl10k bridge-defect images (Hugging Face) — **LIVE** (metadata API); CODEBRIM — **listed only**.

Demo (60 s) Load the MA NBI file and get a ranked list. Each rank is explained by the deck, superstructure, and substructure ratings behind it. Click a bridge to see a sample inspection photo with cracks or corrosion flagged (stretch goal).

Hardest steps (1) Decoding the NBI items correctly (58 deck, 59 superstructure, 60 substructure); one wrong column produces a confidently wrong ranking. (2) A priority score you can defend. (3)

Optional: running a defect model on the CPU over 5–10 images.

Build plan

09:00 Math grad: column-decoding table checked against the FHWA docs and committed to the repo; scoring rule.

10:00 CS1+CS2: parser, scoring, ranked table. CS3: bridge detail view. CS4: defect model on cached sample images.

13:00 Chem major (tester): spot-check 20 bridges against their raw ratings.

15:00 Connect the image flags, or drop to table-only.

16:30 Rehearse the demo (one county). **18:00 freeze**.

Risks / fallback Show the decoding table during the demo; that turns the biggest risk into a point of credibility. Present it as a *maintenance-planning aid*, not a safety verdict. Be ready to say how it differs from FHWA InfoBridge. Without the images it is still a complete demo.

#3 Path-planner classifier — "which algorithm should I use here?"

Track / challenge Robotics — Path Planning Classifier

Scores I3 F5 N4

Problem The challenge text: *"How to choose which path planning algorithm for your environmental needs and compute resources or other constraints."* People pick A* or RRT out of habit. A tool that runs several planners on the same map and recommends one answers the question directly.

Who it helps Robotics students and engineers. Of the five, this is the weakest "real-world problem" and the least specific to MA. It ranks #3 because it is very unlikely to fail.

Data / code

- movingai street benchmark maps: `Boston_0_256` — **LIVE**; `Boston_1/2`, NewYork, Berlin also **LIVE** (enables a multi-map test matrix).
- Boston street centerlines (19,440 segments, real geometry) — **listed only** (data.boston.gov; same Cloudflare caveat).
- PythonRobotics (A*, Dijkstra, RRT, PRM in pure Python) — **LIVE** (repo reachable).

Demo (60 s) Pick a map, a start, and a goal. A*, Dijkstra, RRT, and PRM race side by side, showing path length, nodes expanded, and runtime. The classifier recommends one and says why. Then repeat on the real Boston street graph.

Hardest steps (1) A recommendation rule learned from *many* maps; one map would just make it a lookup table. (2) RRT runs vary and can time out. (3) A shared interface over PythonRobotics scripts, which were written as demos rather than as a library.

Build plan

09:00 Math grad: map features (obstacle density, free-space ratio, size) and the metrics.

09:30 CS1+CS2: wrap the planners behind one interface. CS3: grid UI. CS4: metrics and path overlays.

13:00 Tester: run every map against every planner; fix random seeds and cap iterations.

15:00 Fit the rule from the measured results; add the explanation.

16:30 Rehearse the demo. **18:00 freeze** with precomputed runs.

Risks / fallback Don't claim "optimal"; present it as a guided comparison. Even if the classifier is weak, the side-by-side race works as a demo on its own.

#4 Emergency outreach — who gets missed when a flood is coming

Track / challenge Civic — Stronger Community Emergency Readiness (also fits Climate Resilient communities)
Scores I4 F4 N4
Problem The challenge: *"vulnerable populations can be hardest to reach"*, and information changes quickly. State plans (ResilientMass) work at state level. A small-town emergency manager has no ready list of which block groups need door-to-door outreach for a particular event, or in which languages.
Who it helps Municipal emergency managers and community organizations.
Data

- MassGIS EJ block groups — **LIVE**. OpenFEMA MA disaster declarations — **LIVE**.
- NOAA Boston tide gauge — **LIVE**. NOAA buoy 44013 — **LIVE**. CDC SVI — **listed only**.

Demo (60 s) Simulate a coastal-flood advisory (tide gauge crosses a threshold). The tool ranks the EJ block groups in exposed towns by limited-English % and income, lists the languages outreach must cover, and shows the state's hazard ranking for context. A *replay mode* shows the same view for a past MA FEMA disaster.
Hardest steps (1) Matching block groups to towns to flood exposure without a full GIS stack. (2) A priority rule that stays transparent (no black-box score). (3) Turning a tide threshold into a credible trigger.
Build plan

09:00	Math grad: trigger threshold and priority rule (per-capita normalized).
10:00	CS1+CS2: data adapters. CS3: map and ranked list. CS4: replay mode for past FEMA events.
13:00	Tester: check every rank against the raw block-group values.
15:00	Hazard-context table taken from the MA Climate Change Assessment.
16:30	Rehearse the demo. **18:00 freeze**.

Risks / fallback The tool will **not** write alert text; an official-looking emergency message from an unofficial tool is a liability. It produces a priority list for a person to act on. If the live trigger is unreliable, replay mode is the demo.

#5 Trial finder — MA clinical trials in plain language, by distance

Track / challenge Health — Making Clinical Research More Accessible
Scores I3 F4 N3
Problem The challenge: patients and referring clinicians struggle to find studies; *"eligibility information can be technical, and access may differ by language, geography, and transportation."* ClinicalTrials.gov can be searched, but its eligibility text is dense.
Who it helps Patients and community clinicians outside the Boston research hubs.
Data

- ClinicalTrials.gov API v2, MA + recruiting — **LIVE**.
- Western MA acute hospitals — **LIVE**. EJ block groups — **LIVE**. HRSA shortage areas — **listed only**.

Demo (60 s) Search "type-2 diabetes near Springfield". You get three recruiting MA trials in plain language: what each tests, a one-line eligibility summary, the site, and travel distance. Trials far from a shortage area are flagged.
Hardest steps (1) Plain-language summaries that don't distort eligibility. (2) Geocoding trial sites. (3) Staying on the right side of "discovery, not advice".

Build plan

09:00 Math grad: summary template and the safety boundary.

10:00 CS1+CS2: API query and parsing. CS3: geocoding and distance. CS4: results UI and overlays.

13:00 Chem major (tester): check every summary against its NCT record.

16:30 Rehearse the demo. **18:00 freeze** with the query cached.

Risks / fallback Every summary links to the original record and is labelled as a summary. The tool does not decide eligibility or give medical advice. Novelty is modest; the strength is a clean, safe, data-driven build.

6 Parked ideas (usable backups)

Idea	Track	Status and how it would come back
Climate-risk brief	**Climate**	Best problem, worst feasibility (4+ data sources, a geography join, a composite score). Its core was merged into #4. Bring it back only if cut to *one* hazard (coastal flooding).
MA electric-bill explainer	**Climate**	Strong documented harm: per the AG, competitive-supply customers paid $87.4M more than basic service in Jul 2024–Jun 2025 (energychoicematters.com, 2026-03-31). DPU complaint traces exist. To use portal data it would need the utility-by-town data (**LIVE**), Mass Save data, and ISO-NE.
Permit navigator	**Civic**	Re-tracked Energy → Civic (Digital Government). Named installer quote: "Permitting is the state's biggest blocker" (BU News Service, Oct 2025). Caution: **SolarAPP+ already exists and is free** for towns, so the gap is adoption, not software. An honest civic agent answering from the city's own sources is the angle.
BERDO prep	**Climate**	Boston extended its 2026 deadline and offered free support (boston.gov, 2026-04-15); fines of $150–300/day. The portal now provides the BERDO 2021 dataset. Evidence comes from institutions only.
Chem file validator	**Health**	OpenBabel/RDKit silent-conversion bugs (openbabel #2427, #1049). The team judged the audience too small; it uses no portal data.

7 More ideas for inspiration

These have **not** gone through the full evidence hunt and gates. Each is tied to an official challenge and a portal dataset. The label shows how far each one has been checked:

- **Data checked**: we fetched the main dataset today and it works. The problem itself has not been researched.
- **Idea only**: the dataset is listed on the portal but we have not fetched it.

Use them to spark discussion or to swap in for a shortlist idea. If one gets picked, spend 20 minutes tonight checking that the problem is real and isn't already solved.

Step-free MBTA route checker
Civic: Safer & connected places · **data checked**

Problem Wheelchair users, parents with strollers, and people with injuries find out a station elevator is out only when they arrive. The MBTA publishes elevator-closure alerts, but a normal trip planner doesn't route around them.

Data MBTA v3 API alerts — **LIVE**: on 24 Oct 2026 the feed returned 15 current elevator closures (e.g. Aquarium Elevator 914, State Elevator 803). Rapid-transit stations — **LIVE**.

Demo "Get me from Alewife to Aquarium step-free." The tool reports that Aquarium's elevator is out and suggests getting off at State and taking the accessible exit, with the alert quoted.

Watch out The MBTA's own trip planner may already handle some of this; check tonight. Also needs a list of which stations are accessible.

Robot-demonstration data linter
Robotics: Real-Time Control for Data Quality · **data checked**

Problem The challenge: bad robot-training data (dropped frames, timing drift, poor demonstrations, coverage gaps) is caught too late, after a model has already trained badly. A linter that flags bad episodes *as they're recorded* saves collection time.

Data DROID-100 episode metadata (LeRobot parquet, 138 KB) — **LIVE**. BridgeData episodes index — **listed only**.

Demo Run it over 100 recorded episodes. It flags 7 with frame gaps, 3 where the robot sat idle for most of the episode, and 2 with joint values out of range, then replays one flagged episode's signals.

Watch out Works on metadata and small slices only; don't download full video. Rules must match real failure types (check the LeRobot docs). Very much in the spirit of the challenge, and well suited to a math or stats person.

Swarm search-and-rescue on real Boston streets
Robotics: Swarm Robotics · **data checked**

Problem How many robots does it take to cover an area, and what happens when they lose contact with each other? This is the core question of the swarm challenge.

Data movingai Boston/NewYork/Berlin maps — **LIVE**. PythonRobotics — **LIVE**. RoboSpace browser sim — **listed only**.

Demo Release 1, 3, and 6 simulated robots into a collapsed-street map, each exploring the nearest unexplored area. A graph shows time to full coverage; when communication drops, coverage slows visibly.

Watch out Pure Python on a 2D grid, so feasible. The risk is that it looks like a textbook exercise. Use real street maps and a clear question ("is a 4th robot worth it?").

MassHealth renewal-notice decoder
Health: Keeping Residents Connected to Coverage · **idea only**

Problem Renewal notices are confusing, and renewal rules are changing. About 7,300 MassHealth members lose comprehensive coverage effective 1 Oct 2026 (WGBH, 2026-09-21, verified in Round 2). People who miss a deadline lose coverage.

Data MassHealth federal updates, Medicaid eligibility rules, Census insurance data, EJ block groups for languages — **listed only**.

Demo Paste a *synthetic* renewal notice. The tool explains in plain language what it asks for, the deadline, the documents to gather, and the official place to respond, in Spanish or Haitian Creole.
Watch out Never decide eligibility; always point to the official process. Use only synthetic notices, never real ones. High impact, but has to be built carefully.

Western MA specialty-care travel map
Health: Improving Healthcare Access & Coordination · **data checked**

Problem The challenge points to rural travel distance and limited specialty care. How far does a resident of a hill town actually have to drive to reach an ICU or pediatric bed?
Data Western MA acute hospitals (ICU, med-surg, pediatric beds) — **LIVE**. HRSA shortage areas — **listed only**. MassGIS towns — **listed only**.
Demo Click a town to see drive distance to the nearest ICU, nearest pediatric beds, and whether it is a federal shortage area. Then show the 10 towns farthest from care.
Watch out Licensed beds are not live availability; say so on screen. Risks becoming just a map unless someone specific (e.g. a rural clinic's referral coordinator) uses it.

Ocean forecast vs. fishing-boat reality
Climate: Understanding the ocean / Blue economy · **data checked**

Problem The challenge: measurements from boats, buoys, and models are hard to combine. Fishermen using eMOLT sensors collect bottom temperatures every day; how often does the regional forecast get them wrong?
Data eMOLT bottom temperature — **LIVE** (June 2024 window). NOAA buoy 44013 — **LIVE**. NECOFS FVCOM forecast — **listed only** (NetCDF; harder to load).
Demo Pick a week. Map where measured bottom temperatures differ from the forecast by more than 2 °C, and point out which fishing grounds the model has trouble with.
Watch out Loading NetCDF forecast data is the hard part; budget for it or replace the forecast with a climatology baseline. Highly specific to MA and almost no other teams will pick it.

Honest city-services agent (re-scoped permit navigator)
Civic: Citizen-Friendly Digital Government · **idea only**

Problem The challenge names long PDFs and confusing applications. The portal links Boston's own guide on how a civic AI agent should identify itself and cite city sources.
Data Boston Open Context, civic-agents identity guide, boston.gov pages — **listed only**.
Demo Ask "how do I get a permit to finish my basement?" The agent answers in numbered steps, quotes the city page for each step, and says plainly that it is unofficial.
Watch out Plenty of teams will build chatbots, and the brief warns against generic ones. It only stands out if strict source citation and honesty *are* the feature.

Town clean-energy reality check
Climate: A more reliable clean energy system · **data checked**

Problem Local clean-energy proposals often skip who actually runs the grid in that town and what already exists there. The portal itself says: "name who actually has to deliver the electrons."
Data Electric utility by town — **LIVE**. Renewables at MA landfills — **LIVE**. Transmission lines, ISO-NE — **listed only**.
Demo Pick a town to see its utility, the nearest high-voltage line, and the megawatts already running on its closed landfills.
Watch out Neat but niche; the impact is unclear. Better as a feature inside a larger climate idea.

8 Data portal inventory

The portal lists 166 entries. Below are the ones relevant to our ideas, plus notable others. Status was checked on 24 Oct 2026.

Track	Dataset / resource	Status	Notes / used by
Civic	Boston 311 service requests	BLOCKED	Cloudflare blocks server-side `curl`; use the browser or cache tonight. #1
Civic	MassGIS environmental-justice block groups (2020)	LIVE	Minority %, limited-English %, median income. #1, #4, #5
Civic	OpenFEMA disaster declarations (MA)	LIVE	Title, date, incident type. #4
Civic	MBTA v3 API (routes, alerts, stations)	LIVE	Not used yet; useful for transit ideas.
Civic	Boston permits; Boston neighbourhood ACS estimates	listed only	Same Cloudflare caveat. Permit navigator.
Civic	Boston Open Context; civic-agents identity & auth guide	listed only	How to build a civic AI agent that cites sources and identifies itself.
Civic	Cambridge / Somerville bike and pedestrian counts; Vision Zero crashes	listed only	Safer-places ideas.
Civic	College Scorecard, IPEDS, O*NET, BLS wages, mass.edu data center	listed only	Former "education" ideas → Civic pathways.
Climate	NOAA Boston tide gauge (8443970)	LIVE	Hourly water level. #4
Climate	NOAA buoy 44013 (Boston)	LIVE	Waves, wind, water temperature.
Climate	USGS Charles River at Waltham (flow)	LIVE	River-flood context.
Climate	eMOLT bottom temperature (ERDDAP)	LIVE	Returns data for Jun 2024; was empty for the first window tried (Aug 2024). Check the date range.
Climate	Electric utility by town (351 rows)	LIVE	Bill explainer; grid siting.
Climate	Renewables at MA landfills	LIVE	Grid siting.
Climate	Transmission lines; ISO-NE Express; Mass Save data	listed only	Grid and buildings ideas.
Climate	Boston BERDO 2021 reporting	listed only	BERDO prep (Cloudflare caveat).
Climate	MA Climate Change Assessment; ResilientMass; NOAA Digital Coast	listed only	Background documents. #4

Track	Dataset / resource	Status	Notes / used by
Climate	FIShBOT, NECOFS forecast, NASA PACE, WHOI HABhub, NE Ocean Data Portal	listed only	Ocean and blue-economy challenges (no idea yet).
Health	ClinicalTrials.gov API v2 (MA, recruiting)	LIVE	#5
Health	Western MA acute hospitals (ICU, beds)	LIVE	#5; rural access.
Health	CDC PLACES (MA)	LIVE	County health burden.
Health	CMS hospital ratings and readmissions (MA)	listed only	Access and coordination.
Health	Synthea synthetic patients; HL7 FHIR; CMS care-update template	listed only	Safe prototyping without real patient data.
Health	WHOOP developer API; PhysioNet wearable dataset	listed only	Wearables challenge. The WHOOP API needs user consent; PhysioNet may have access rules.
Health	MassHealth federal updates; Medicaid eligibility; Census insurance; CDC SVI	listed only	Coverage challenge (no idea yet).
Health	Open Targets, ChEMBL, GEO, Protein Atlas, PubChem, openFDA	listed only	Biotech challenge.
Robotics	National Bridge Inventory, MA (`MA22.txt`)	LIVE	#2
Robotics	dacl10k bridge defects (Hugging Face)	LIVE	#2 stretch goal; check the licence before showing images.
Robotics	CODEBRIM; FHWA InfoBridge; SNBI docs	listed only	#2
Robotics	movingai street maps (Boston_0/1/2, NewYork, Berlin)	LIVE	#3
Robotics	PythonRobotics	LIVE	#3
Robotics	RoboSpace (browser simulator, 70+ robots)	listed only	No install; option for swarm / robo-gym ideas.
Robotics	MuJoCo, ManiSkill, Gazebo, ROS 2 / Nav2, OR-Tools	listed only	Gazebo/ROS take time to install; budget for it.
Robotics	Isaac Sim / Isaac Lab	not usable	Needs an NVIDIA RTX GPU. Not usable on our laptops.
Robotics	Open X-Embodiment index; DROID-100; ARMBench; GraspNet	listed only	Data-quality challenge. The full datasets are huge; use slices.

9 Kill and drop ledger (all rounds)

Most earlier ideas were dropped because the space was crowded, not because of track fit, so the corrected tracks bring back very few. The data-broker idea (s7) now fits Civic, but it stays dropped: DeleteMe, Incogni, Optery, and PrivacyBee already cover it.

Idea	Round	Outcome	Reason
Mass Save rebate tracker	2→3	dropped	Was #1 in Round 2. Real victims (WBZ I-Team, 2021-12-10 and 2023-12-18; $7.7M fraud suit, Globe 2026-09-22), but uses no portal data, and contractor-side rebate tools exist (Resolv, Instafill, Homewise).
FRC mentor logging	2→3	dropped	Real recurring Chief Delphi complaints (2019–2026), but outside the robotics track's physical-AI focus and uses no portal data.
FRC inspection self-check	2→3	dropped	Same; FIRST owns this workflow and is modernizing it.
Lab-protocol scaling	1→3	dropped	No Education track; fits no real track.
Caption QA for lectures	1→2	dropped	3Play Media (2,000+ universities) and Verbit already cover it.
Student-feedback report	1→2	dropped	Many vendors; no Education track.
Manim animation helper	1	dropped	No trace of teacher demand; no-code competitors.
Attention blocker	1	dropped	Freedom, Cold Turkey, Forest, Opal, GoGuardian.
CHW paper-to-DHIS2	1	dropped	KoBo, ODK, CommCare are mature.
Vaccine cold chain	1	dropped	Nexleaf (43 countries); needs hardware.
Protein-structure gaps	1	dropped	AlphaFold DB (260M+) fills the gap.
Worksheet → widget	1	dropped	Desmos/Amplify, free and widely used.
Rubric-consistency checker	1	dropped	Only second-hand evidence.
EMS documentation	1	dropped	ESO/ZOLL; NEMSIS data access needs approval.
Adult-ed reporting	1	dropped	No first-person trace.
Data-broker opt-out (s7)	1, re-checked 3	dropped	Now fits Civic, but 4+ funded competitors; MA Data Privacy Act still in conference committee.
Robotics sim curricula	1	dropped	Premise wrong: MA community colleges got hardware grants; Webots course labs exist.
Food pantry ops	2	dropped	PantrySoft, Food Rescue Hero, and others at every price point.

Idea	Round	Outcome	Reason
Interpreter scheduling	2	dropped	Vendors are donating AI interpretation minutes to free clinics.
Free-clinic ops	2	dropped	Practice Fusion free EHR via an NAFC partnership; ClinicHub.
Science-fair coordination	2	dropped	zFairs (used by the MA Science & Engineering Fair); Scilympiad.
Substitute coordination	2	dropped	Frontline, PowerSchool, Tyler, Red Rover.
Heat-pump quoting	2	dropped	No contractor complaint found; quoting software is crowded.
Condo EV charger approvals	2	dropped	Law verified (c.183A s.10A), first-person traces thin.

10 Independent audit findings

A separate reviewer agent read the cards without our reasoning and critiqued them. Its main points, and what we changed:

Idea	Auditor's concern	Change made
311 triage	Cloudflare must be planned for, not a fallback; the equity pattern may just reflect report volume	Cached slice is the main plan; per-capita normalization; choose the neighbourhood from the data
Bridge triage	A wrong NBI column means a confidently wrong ranking; InfoBridge overlap	Committed decoding table shown in the demo; framed as a planning aid
Path planner	One map makes the "classifier" a lookup; weakest problem	Multi-map matrix (confirmed: more maps available); real Boston street graph
Emergency outreach	Writing alert text is a liability; near-duplicate of the climate brief	Alert writing removed; climate brief merged in and parked
Climate brief	Too many integrations for 11 h	Parked

Auditor's ranking: 311 triage, bridge triage, path planner, emergency outreach, climate brief (cut). We kept that order and filled slot 5 with the trial finder so the list covers all four tracks.

11 Open questions and tonight's checklist

1. **Read the rules in a browser**: the Notion page (`https://rich-uranium-d4e.notion.site/b836a9c5ce974d32a09bbec1bd38f928`) and the event page (`https://massaicoalition.com/hackachusetts/`). We could not read either. Check the judging criteria, team-size limits, and whether ideas must name a challenge.

2. **Pick one idea.** Recommended: #1 if someone can download the 311 slice tonight; otherwise #2. Keep #3 as the backup.
3. **311**: open the data.boston.gov 311 dataset in a browser and export a CSV for one or two neighbourhoods. Look for an SLA-miss pattern before committing.
4. **Bridges**: download `MA22.txt` and find the FHWA coding guide for items 58–60.
5. **Path planner**: `git clone` PythonRobotics and run one A* and one RRT script on each laptop.
6. **Everyone**: confirm the LLM coding agent works on each laptop (one seat each, or shared?).
7. **Track strategy**: Civic is the most open track (#1 and #4). Concentrate there, or spread across tracks?
8. **Not yet researched**: see Section 7 for starting points. Still untouched: wearables, manufacturing automation, robo-gym.